Insightify 6.0

The Muggles

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Problem Statement

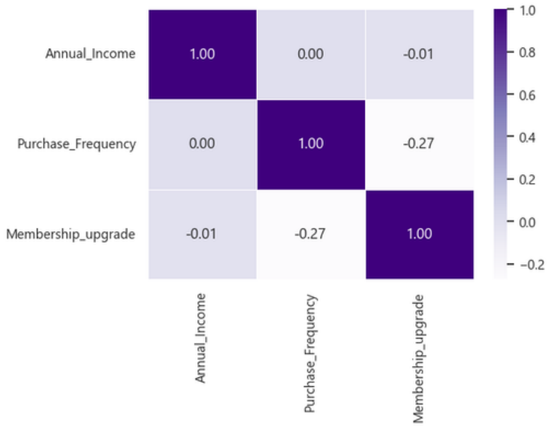
CraveConnect operates a tiered membership program **(Basic → Silver → Gold → Platinum)** designed to increase customer retention and revenue. We need to identify **"upgrade-ready" customers.**

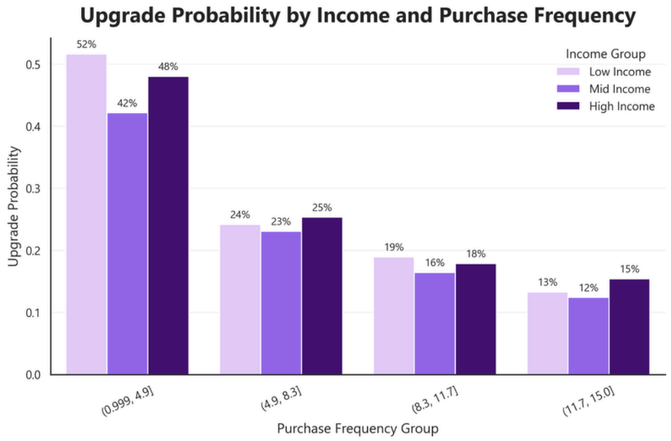
Demographic profile of a typical users (Mean)

Membership Level	Age	Annual Income	Purchase Frequency	Avg Order Value	Spending Score	Total Cuisines Tried
Basic	43.85	61,224.58	8.91	60.49	57.28	5.49
Gold	43.47	59,110.21	8.16	60.7	50.38	5.45
Platinum	43.57	60,687.84	8.22	60.03	50.3	5.38
Silver	43.5	59,613.92	7.81	59.3	50.01	5.38

Demographics **do not strongly differentiate membership tiers**, so predictive modeling should focus on behavioral signals.

The correlation between Annual_Income, Purchase_Frequency, and the likelihood of an upgrade

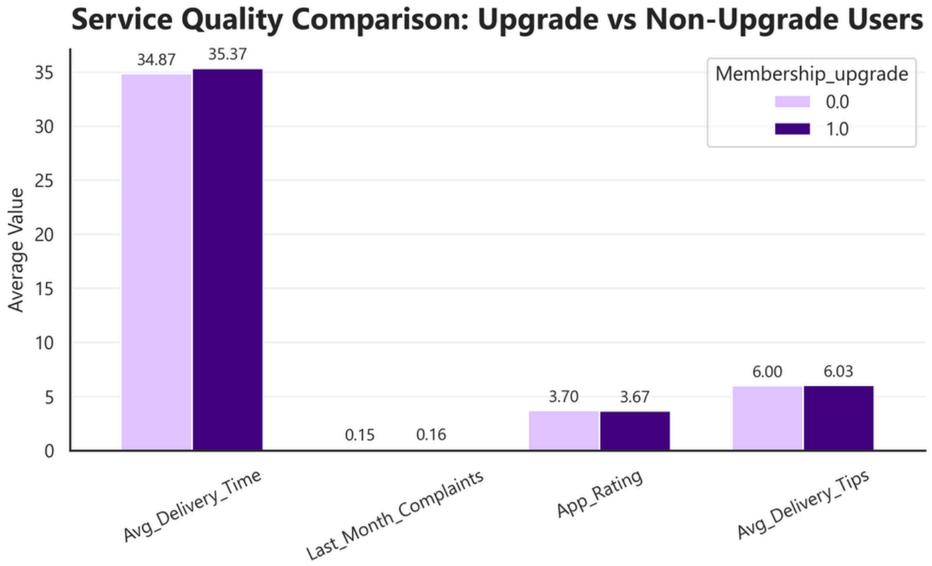




- **Purchase frequency** shows a **stronger correlation (-0.27)** with **membership upgrades** compared to **annual income (-0.01)**, indicating that **platform engagement plays a larger role in driving premium adoption**.
- **Annual income exhibits a weak relationship with upgrade likelihood**, suggesting that **customers across income levels may consider upgrading.**

How service quality impacts a user's willingness to upgrade.

Membership Upgrade	Avg Delivery Time	Last Month Complaints	App Rating	Avg Delivery Tips
0 (No Upgrade)	34.87	0.15	3.7	6
1 (Upgraded)	35.37	0.16	3.67	6.03



Service Experience Has Limited Influence on Upgrade Behavior

- **Delivery time** and **complaint rates show minimal differences** between upgrade **(35.37)** and non-upgrade users **(34.87)**, suggesting service issues are not a major driver of membership upgrades.
- **Customer ratings and delivery tips remain relatively consistent across both groups (3.70 vs 3.67)**, indicating overall satisfaction levels are similar.
- **Last Month Complaints and Avg Delivery Tips show a similar pattern**.
- This shows that **upgrade behavior appears to be relatively less driven by services quality**.

Data Cleaning

- **Removing missing target values:** Rows where the Membership_upgrade variable was missing were excluded to ensure valid training data. **(200 NaN Values)**
- **Dropping non-predictive identifiers**: Columns such as **CustomerID** and **Name** were removed as they do not provide predictive value.
- **Avoiding data leakage**: The variable **Membership_Level** was excluded from the feature set because it could trivially indicate upgrade behavior.

The dataset contains **more non-upgrade customers** than **upgrade customers**, creating class imbalance.
To address this, the Random Forest model uses:
class_weight = "balanced"

Class	Ratio
0 (Not Upgraded)	0.74
1 (Upgraded)	0.26

Data Preprocessing

The dataset contains both numerical and categorical features.

Data Type	Method
Numerical variables	StandardScaler
Categorical variables	OneHotEncoder

Feature Engineering

To improve model performance and capture behavioral insights, several engineered features were created.

Feature Name	Formula	Meaning
Spend_Efficiency	Spending_Score / (Purchase_Frequency + 1)	Spending relative to usage frequency
Complaint_Rate	Last_Month_Complaints / (Purchase_Frequency + 1)	Complaints normalized by usage
Cuisine_Diversity	Total_Cuisines_Tried / 8	Exploration of different cuisines
Delivery_Pain_Index	(Avg_Delivery_Time / 60) + Last_Month_Complaints	Combined delivery friction indicator
Tip_to_Order_Ratio	Avg_Delivery_Tips / (Avg_Order_Value + 1)	Tipping behavior relative to order value

Data Splitting

The dataset was divided into:
Training samples: (1440, 19) (80%)
Testing samples: (360, 19) (20%)
Stratified sampling was used to ensure that the proportion of upgrade and non-upgrade customers remained consistent across both sets.
This allows reliable evaluation of model performance on unseen data.

Model Selection

For predicting, two models were used:
Random Forest
Random Forest was chosen because it can capture complex, non-linear relationships and interactions between customer behavior features (such as spending patterns and purchase frequency), leading to more accurate upgrade predictions.

Logistic regression
Logistic Regression was chosen because the task is a binary classification problem (upgrade vs. no upgrade) and it provides a simple, interpretable baseline to understand how customer features influence upgrade probability.

Model Evaluation

The following was the output on evaluating the models

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	PR-AUC
Logistic Regression	0.65	0.39	**0.613**	0.475	0.70	**0.573**
Random Forest	0.817	0.89	**0.333**	0.484	0.71	**0.603**

Random Forest has much **higher accuracy (81.7%)** and **precision (88.6%)**, meaning its **positive predictions are more reliable**.

Logistic Regression has **higher recall (61.3%)**, meaning **it identifies more actual upgrade cases.**

ROC-AUC and **PR-AUC** are similar, suggesting both models have comparable overall discrimination ability, but **Random Forest is more consistent in predicting upgrades.**

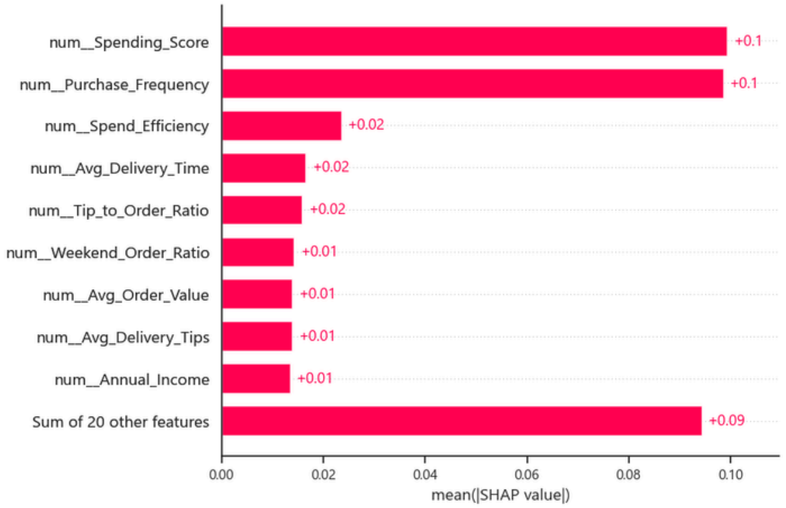
The trained **Random Forest model (rf_model)** was then used to **calculate the probability that each customer will upgrade their membership.**

Customers whose predicted upgrade probability is **greater than 0.7 (70%)** were **selected (314).**

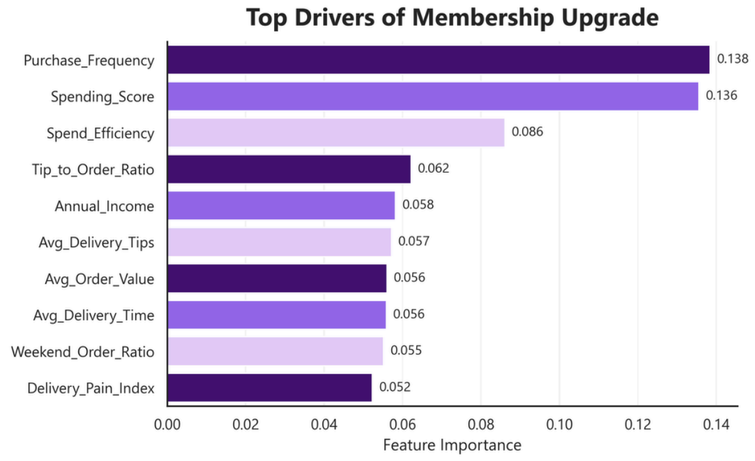
Raw Dataset → Data Cleaning & Feature Engineering → Train-Test Split (80/20) → Model Training (Logistic Regression & Random Forest) → Model Evaluation (PR-AUC, ROC-AUC, F1) → Best Model Selected (Random Forest)

Top 3 Drivers of membership upgrade

Using **SHAP** to interpret the predictions of the Random Forest model.



Using **Random Forest feature importance** to interpret the predictions of the Random Forest model.



Who is our Target Customer?

Attribute	Mean Value	Interpretation
Age	43.48	Mid-career professional with stable lifestyle
Annual Income	59,371.76	Moderate to high purchasing power
Purchase Frequency	5.6	Regular platform user
Average Order Value	60.03	Willing to spend on quality meals
Spending Score	31.96	Moderate engagement with spending patterns
Total Cuisines Tried	5.25	Enjoys exploring food variety
Weekend Order Ratio	0.51	Orders equally on weekdays and weekends

Key Behavioural Characteristics	Busy Professional Frequent Platform User Cuisine Explorer Value-Seeking Customer

Persona Story - Rahul

Age: 43 | Mid-Career Professional | Busy Urban Lifestyle

Rahul is a **43-year-old professional** who frequently uses **CraveConnect** to simplify his busy routine. With an **annual income of around $59K, he orders food 5–6 times a month, spending about $60 per order.** He enjoys **exploring different cuisines** and regularly **tries new restaurants through the app**.

"Weekend Explorer Deals"

CraveConnect targets customers like Rajiv with a personalized upgrade campaign with curated recommendations could highlight trending cuisines or limited-time restaurant collaborations.

The Strategy

Although Rahul is an active user, he still uses the Basic membership tier. Since he orders frequently and values convenience, a Gold membership offering reduced delivery fees and priority service would provide clear value.

CraveConnect introduces **"Weekend Explorer Deals"** for customers like Rahul, since his **ordering behavior is balanced between weekdays and weekends.** These curated recommendations could **highlight trending cuisines or limited-time restaurant collaborations, giving Rahul the feeling that he is part of an exclusive food community rather than just a regular user.**

A targeted message such as:
"Rahul, since you enjoy exploring new cuisines, try these chef-recommended restaurants today with free delivery through your Gold trial."

can encourage him to **experience the benefits firsthand.**

As **Rahul experiences smoother deliveries, curated restaurant recommendations, and reduced fees, the Gold membership starts to feel like a natural upgrade rather than an extra cost.** By the end of the trial period, Rahul chooses to upgrade to a **Gold membership.**

For **CraveConnect**, this strategy successfully converts a **consistent but moderately engaged** user into a **long-term premium member.**

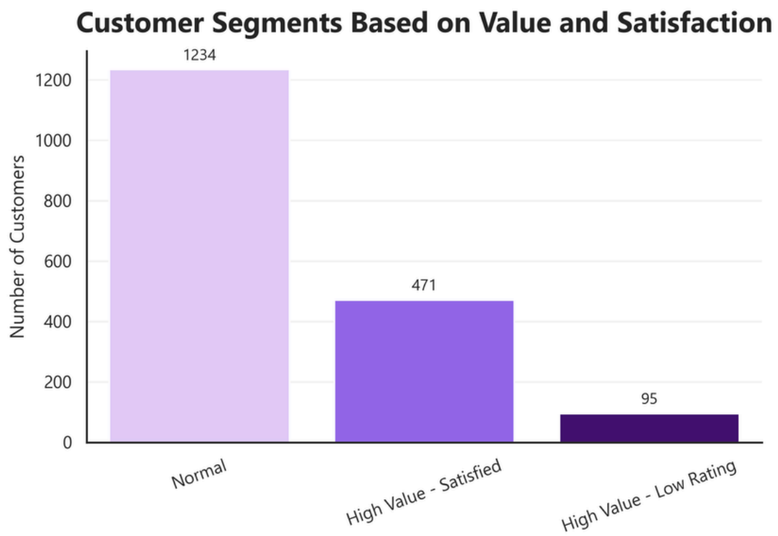
"High-Value/Low-Rating" Users

Customers with a **Spending Score above 70** but an **App Rating below 3** represent a **high-value yet dissatisfied segment.** Despite their **strong spending behavior and frequent orders**, their **low satisfaction levels** suggest a **risk of churn**.

Metric	High Value – Low Rating Customers	Other Customers
Purchase Frequency	8.44	8.20
Spending Score	85.26	49.86
Spend Efficiency	11.65	7.14

"High-Value/Low-Rating" Users: 95

Although **High Value – Low Rating customers behave similarly to other users** in terms of **order frequency**, their **spending score (85.26) and spend efficiency (11.65)** is **dramatically higher**, indicating **deeper engagement** and **greater lifetime value**.



Although the **High Value – Low Rating** segment represents only about **95 (5.28%)** of customers, they are strategically important because they generate significant revenue but show signs of dissatisfaction. This makes them a priority for targeted retention strategies.

Persona Story - Neha

Age: 36 | Strategy Consultant | Food Delivery Power User

Neha lives a fast-paced urban life **where food delivery isn't a luxury, it's a daily convenience**. After long days at work, she often opens **CraveConnect** to unwind, trying out new restaurants and cuisines. With a **Spending Score above 70**, Neha is one of the **platform's most valuable customers**, ordering frequently and spending significantly on meals.
But recently, her **experience has changed**. L**ate deliveries, cold food, and inconsistent service have started to disrupt what used to be a seamless routine.** Despite still ordering regularly, Neha's frustration shows through her low app rating.
Neha hasn't left the platform yet, but **she's starting to wonder if another app might serve her better.**

Key Characteristics	High spending power Frequent platform user Food discovery enthusiast Service-sensitive customer

The Idea

"Win Back the Food Lover" Campaign

Instead of treating Neha like just another dissatisfied customer, CraveConnect treats her like a VIP foodie whose experience matters.

The Strategy

One evening, Neha opens the app and sees a personalized message:

"Neha, we noticed you love discovering great food. Let us make it better for you, enjoy a Gold/Platinum Experience Pass for 30 days, including priority delivery and chef-curated restaurant picks."

The app now highlights exclusive restaurant recommendations, faster delivery options, and limited-time deals from trending restaurants.

What used to be frustrating becomes exciting again Neha feels like she has access to a premium food discovery experience rather than just another delivery app.

By turning a negative experience into a personalized recovery journey, **CraveConnect** achieves two goals:

1. Neha **feels valued as a customer** rather than just another user.
2. The **upgraded membership** trial allows her to experience the real value of premium service.

What started as a **retention strategy** becomes an opportunity to **convert a dissatisfied high-value customer into a loyal upgraded member**.

Thank you

Made By

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